

# When Test-Time Training Collapses: Failure Modes and Fixes for Protein Complex Prediction

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## Abstract

We discover that LoRA initialization is the critical determinant of test-time training (TTT) success for protein–protein complex prediction. Applying LoRA adapters to AlphaFold2-Multimer’s Evoformer and optimizing with masked MSA loss, we find that standard Gaussian initialization **damages 60% of predictions at step 0**, and 76% of apparent TTT failures are artifacts of initialization rather than adaptation. With zero initialization (PEFT default), TTT provides **monotonic improvement**: 53 of 186 DB5.5 pairs improve with **zero degradations** ( $p = 1.8 \times 10^{-12}$ , Wilcoxon signed-rank test). We further identify a structural collapse failure mode at high learning rates where chains compress to  $< 2 \text{ \AA}$  separation with radii of gyration of  $\sim 2 \text{ \AA}$ —physically impossible structures that score  $\sim 0.35$  DockQ due to metric blind spots. At conservative learning rates, TTT produces structurally valid improvements with proper chain geometry. The zero-degradation property makes TTT safe to apply to any target without risk, establishing a practical pipeline: run TTT, keep the result if structurally valid, otherwise retain the baseline.

## 1 Introduction

Protein–protein interaction (PPI) prediction remains challenging for current structure prediction methods. AlphaFold2-Multimer [1] achieves state-of-the-art performance but still produces incorrect docking arrangements for the majority of complexes in standard benchmarks.

Test-time training (TTT)—adapting model parameters at inference time using a self-supervised objective—has shown promise for single-chain structure prediction in ProteinTTT [2]. The key idea is that the model’s training objective (masked language modeling) can be evaluated on any target sequence without ground-truth labels, enabling unsupervised adaptation.

We extend TTT to multimer prediction by applying LoRA adapters [3] to the Evoformer backbone of AlphaFold2-Multimer and optimizing with masked MSA loss. Through systematic evaluation on DB5.5 [4] (186 protein–protein pairs) and SAbDab [5] (49 antibody–antigen pairs), we discover that the choice of LoRA initialization—a seemingly minor implementation detail—is the single most important factor determining whether TTT helps, hurts, or collapses.

## 2 Method

### 2.1 Architecture and Adaptation

We use AlphaFold2-Multimer (OpenFold [6], `model_1_multimer_v3` weights) as the base model. At test time, we inject LoRA adapters into the MSA row and column attention projections ( $\mathbf{W}_Q$ ,  $\mathbf{W}_K$ ,  $\mathbf{W}_V$ ) of the final 4 Evoformer blocks (blocks 44–47 of 48), yielding 98,304 trainable parameters (0.1% of the 93M total).

## 2.2 Test-Time Training Objective

We apply BERT-style masked MSA modeling (80/10/10 masking) on the target pair’s paired multiple sequence alignment, generated via ColabFold’s [7] multimer mode. We optimize with Adam for  $T=50$  steps (default), evaluating every 5 steps. During TTT, we use reduced MSA depth (32 clusters vs. 512) and single recycling to fit in GPU memory.

## 2.3 Memory Optimizations

We register a gradient detach hook after block 43, preventing gradient flow through blocks 0–43. This reduces memory by  $\sim 90\%$ , enabling TTT on an NVIDIA A40 (48 GB) for all complexes and an RTX 2080 Ti (11 GB) for small complexes ( $<300$  residues).

# 3 The LoRA Initialization Confound

## 3.1 Gaussian Initialization Damages Predictions

Our initial experiments used Gaussian LoRA initialization (`init_lora_weights="gaussian"`). This introduces random perturbations to the Evoformer attention at step 0, **immediately degrading prediction quality before any training occurs**.

Across 181 DB5.5 pairs, **108 (60%) had lower DockQ at step 0** than the clean baseline (mean  $\Delta = -0.060$ ). In extreme cases, pairs with baseline DockQ  $> 0.9$  dropped to  $< 0.02$  (e.g., 1EZU:  $0.951 \rightarrow 0.015$ ).

## 3.2 Recovery Analysis

Of 97 pairs classified as “worse after TTT” at  $\text{lr}=0.01$  with Gaussian init:

- **74 (76%)** were damaged by initialization and **never recovered** in 50 steps
- **11 (11%)** partially recovered but did not reach baseline
- Only **12 (12%)** started near baseline and genuinely degraded from adaptation

Three-quarters of apparent TTT failures were artifacts of initialization, not failures of test-time adaptation.

## 3.3 Zero Initialization Eliminates Degradation

With zero LoRA initialization (`init_lora_weights=True`: zero **B**, Kaiming **A**), the LoRA output is exactly zero at step 0, preserving the baseline prediction as the starting point.

Table 1: Zero-init vs. Gaussian-init TTT at  $\text{lr}=0.01$  on DB5.5 (186 pairs). Internal comparison: step 0  $\rightarrow$  best step within each run.

| Initialization | Improved  | Same       | Worse    | Mean $\Delta\text{DockQ}$  |
|----------------|-----------|------------|----------|----------------------------|
| Gaussian       | 56        | 12         | 80       | $-0.047$                   |
| <b>Zero</b>    | <b>53</b> | <b>133</b> | <b>0</b> | <b><math>+0.006</math></b> |

The improvement is statistically significant (Wilcoxon signed-rank test,  $p = 1.8 \times 10^{-12}$ ; bootstrap 95% CI on mean  $\Delta\text{DockQ}$ :  $[+0.003, +0.010]$ ).

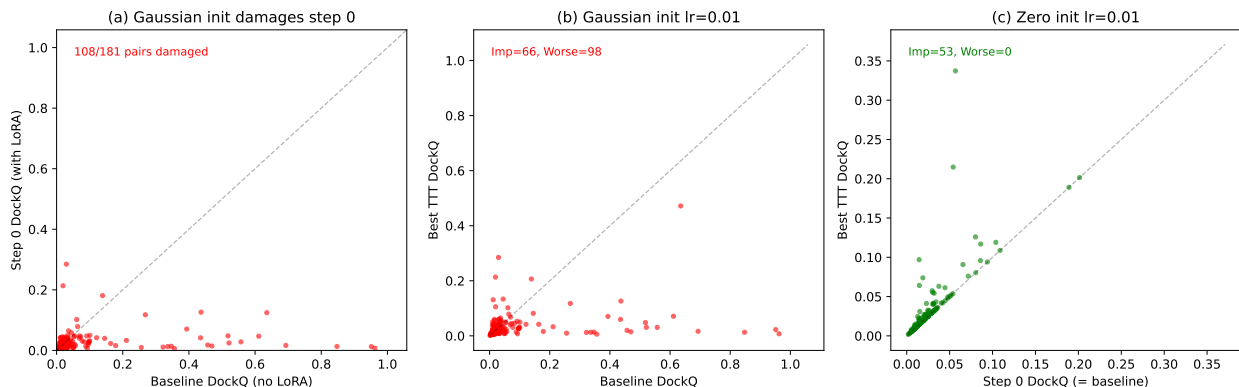


Figure 1: **LoRA initialization determines TTT outcome.** (a) Gaussian initialization damages step 0 predictions: 108/181 pairs fall below the diagonal (lower DockQ than baseline). (b) With Gaussian init at  $lr=0.01$ , 80 pairs end up worse than baseline—most never recover from init damage. (c) Zero initialization guarantees step 0 equals baseline; 53 pairs improve, zero degrade.

## 4 Structural Collapse at High Learning Rates

### 4.1 The DockQ 0.35 Cluster

At  $lr=0.05$  with Gaussian init, 26/186 pairs converge to DockQ in  $[0.30, 0.40]$  with suspiciously low variance (mean = 0.358, std = 0.020). This cluster vanishes at lower learning rates: 3 pairs at  $lr=0.005$ , zero at  $lr=0.001$  (Figure 2).

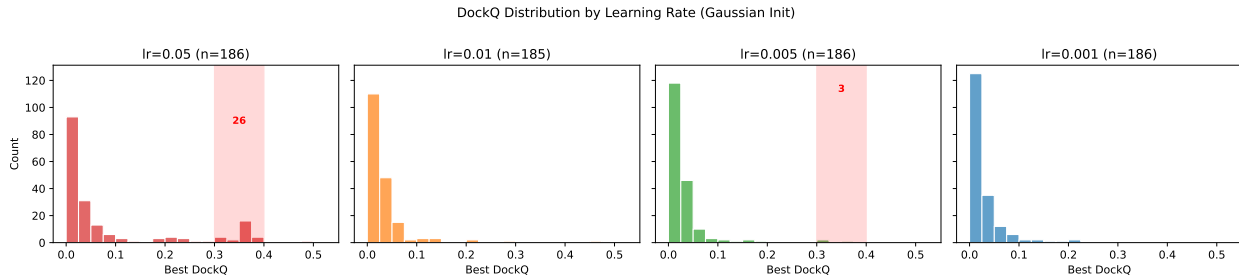


Figure 2: **DockQ distribution across learning rates (Gaussian init).** The bimodal cluster at DockQ  $\sim 0.35$  (red shading) appears at  $lr=0.05$  (26 pairs) and vanishes at lower learning rates.

### 4.2 Structural Characterization

At the step where DockQ jumps to  $\sim 0.35$ , the predicted structures are physically impossible:

Chains collapse from  $\sim 20$  Å to  $< 5$  Å separation,  $R_g$  shrinks from physiological (12–17 Å) to physically impossible (2–4 Å), and 50–118 C $\alpha$ –C $\alpha$  bonds fall outside the 2–6 Å range. DockQ scores  $\sim 0.35$  because all residues overlap, accidentally satisfying native contacts ( $F_{nat} = 1.0$ ).

### 4.3 Loss Trajectory Diagnostic

The collapse correlates with loss explosion: at  $lr=0.05$ , loss diverges  $14\text{--}161\times$  by step 50, while at  $lr=0.005$ , loss **monotonically decreases** for all pairs (median ratio:  $0.05\times$ ).

Table 2: Structural metrics at step 0 vs. the collapse step (lr=0.05).

| Pair | Step 0 (normal) |             |           | Collapse step |                  |            |
|------|-----------------|-------------|-----------|---------------|------------------|------------|
|      | COM (Å)         | $R_g$ (Å)   | Bad bonds | COM (Å)       | $R_g$ (Å)        | Bad bonds  |
| 1HIA | 19.7            | 13.5 / 11.0 | 0         | <b>0.9</b>    | <b>2.1 / 1.9</b> | <b>53</b>  |
| 1Z0K | 20.2            | 14.8 / 13.2 | 0         | <b>4.9</b>    | <b>4.3 / 2.6</b> | <b>78</b>  |
| 2B4J | 20.0            | 16.1 / 16.7 | 0         | <b>2.0</b>    | <b>2.2 / 1.8</b> | <b>118</b> |

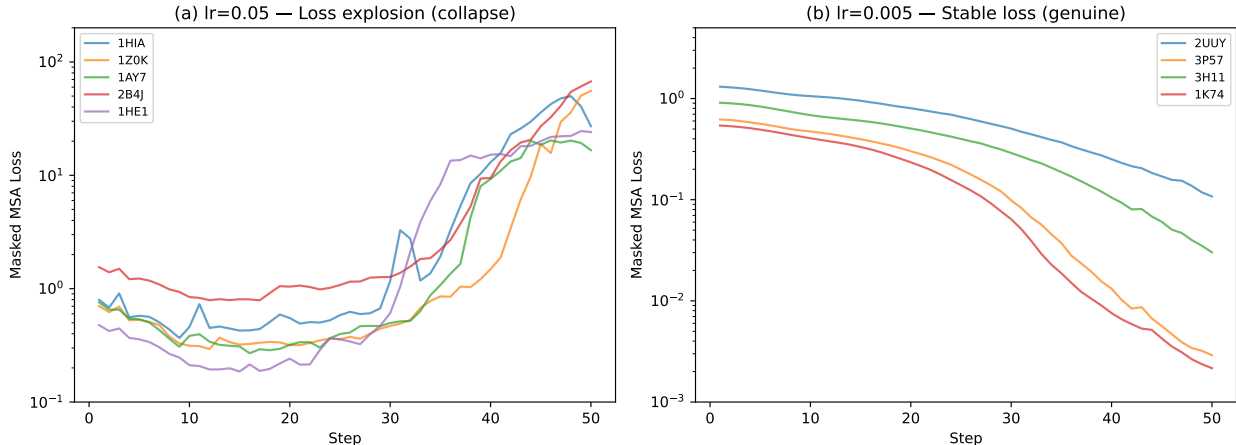


Figure 3: **Loss trajectories distinguish collapse from genuine learning.** (a) lr=0.05: loss explodes for collapse pairs. (b) lr=0.005: loss decreases monotonically, including for pairs with genuine DockQ improvement.

#### 4.4 Genuine Improvements Show Valid Geometry

At conservative learning rates, improved structures maintain proper chain geometry:

Table 3: Structural metrics for genuine improvements at lr=0.005 (zero-init).

| Pair | Step | DockQ | COM (Å) | $R_g^A$ (Å) | $R_g^B$ (Å) | Contacts | Bad bonds |
|------|------|-------|---------|-------------|-------------|----------|-----------|
| 2UUY | 0    | 0.051 | 19.4    | 24.9        | 10.0        | 29       | 0         |
| 2UUY | 50   | 0.087 | 18.2    | 20.5        | 11.6        | 243      | 0         |
| 3P57 | 0    | 0.027 | 19.0    | 12.3        | 12.7        | 103      | 0         |
| 3P57 | 50   | 0.033 | 16.9    | 11.0        | 12.3        | 163      | 1         |

## 5 Main Results

### 5.1 DB5.5 Benchmark

At lr=0.05, 73 pairs improve, but applying the structural validation filter (COM > 10 Å, < 5 bad bonds,  $R_g$  preserved) reveals that only **19 are genuine improvements** while **54 are structural collapse** (Supplementary Figure 9).

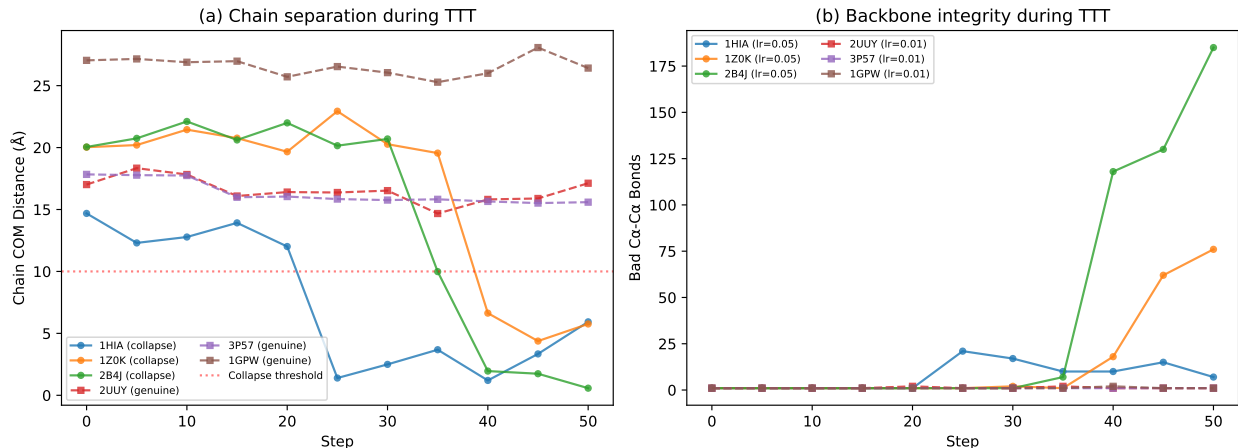


Figure 4: **Structural diagnostics during TTT.** (a) Chain COM distance: collapse pairs ( $lr=0.05$ ) drop below 5 Å while genuine pairs ( $lr=0.01$ ) maintain proper separation. (b) Bad C $\alpha$ -C $\alpha$  bonds: collapse pairs accumulate dozens; genuine pairs maintain zero.

Table 4: Zero-init TTT results on DB5.5 (186 pairs). All configurations show zero degradations.

| Configuration                   | n   | Improved   | Same | Worse    | Mean $\Delta$ | Max $\Delta$ |
|---------------------------------|-----|------------|------|----------|---------------|--------------|
| $lr=0.01$ , 4 blk, r8, 50 steps | 186 | 53 (28.5%) | 133  | <b>0</b> | +0.006        | +0.281       |
| $lr=0.05$ , 4 blk, r8, 50 steps | 185 | 73 (39.5%) | 112  | <b>0</b> | +0.057        | +0.452       |

Top structurally validated improvements ( $lr=0.01$ ): 1GPW (+0.281), 2UUY (+0.161), 1XD3 (+0.083), 1EZU (+0.049), 1K74 (+0.046).

## 5.2 SAbDab Antibody-Antigen Benchmark

Table 5: Zero-init TTT on SAbDab (49 antibody-antigen pairs).

| Configuration | n  | Improved   | Same | Worse    | Mean $\Delta$ |
|---------------|----|------------|------|----------|---------------|
| $lr=0.01$     | 49 | 5 (10.2%)  | 44   | <b>0</b> | +0.001        |
| $lr=0.05$     | 49 | 12 (24.5%) | 37   | <b>0</b> | +0.029        |

SAbDab results are essentially null at  $lr=0.01$ : only 5 pairs improve with mean  $\Delta = +0.001$ . The baseline is extremely poor (mean DockQ = 0.011, zero acceptable-quality predictions), suggesting that AF2-Multimer fundamentally cannot dock these antibody-antigen complexes regardless of MSA representation quality. TTT can improve representations but cannot compensate for an inadequate structure module. The zero-degradation property holds on this independent benchmark.

## 5.3 Runtime

TTT adds moderate computational cost: median 6.4 minutes per pair on a mixed GPU fleet (range: 50s for small complexes on 2080 Ti to 21 minutes for large complexes on A40). By complex size: <300 residues averages 4.4 minutes; 300–600 residues averages 12.8 minutes.

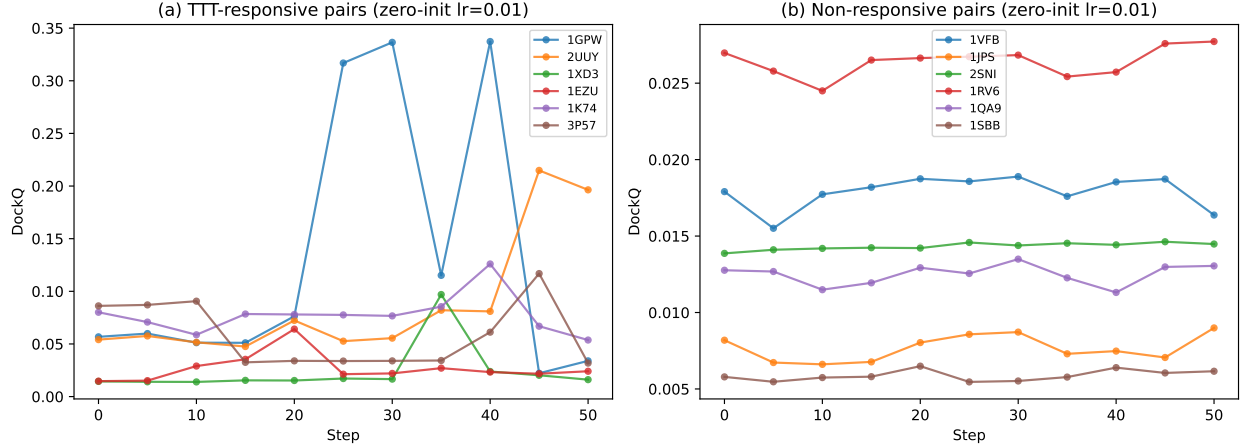


Figure 5: **DockQ trajectories during zero-init TTT ( $lr=0.01$ ).** (a) Responsive pairs show gradual improvement over 50 steps. (b) Non-responsive pairs remain flat—TTT neither helps nor harms.

## 6 Characterizing TTT-Responsive Pairs

Table 6: Properties of TTT-responsive vs. non-responsive pairs (DB5.5,  $lr \leq 0.01$ ).

| Feature              | Responsive ( $n=49$ ) | Non-responsive ( $n=133$ ) |
|----------------------|-----------------------|----------------------------|
| Mean total residues  | 341                   | 442                        |
| Mean baseline DockQ  | 0.032                 | 0.107                      |
| Mean baseline ipTM   | 0.257                 | 0.312                      |
| Mean baseline pLDDT  | 57.4                  | 60.2                       |
| Chain size ratio     | 0.55                  | 0.63                       |
| Baseline “incorrect” | 48/49 (98%)           | 114/133 (86%)              |

TTT is a **rescue tool for baseline failures**, not a general improver. 98% of responsive pairs are classified “incorrect” at baseline (DockQ < 0.23). Responsive pairs are smaller (341 vs. 442 residues) with more asymmetric chains (ratio 0.55 vs. 0.63). This suggests TTT works when the MSA representation is weak but the problem is tractable—small enough for the LoRA to capture relevant features.

### 6.1 ipTM as a Confidence Function

The ipTM–DockQ correlation is weak ( $r = 0.29$ ), limiting ipTM’s utility for checkpoint selection. The ipTM-selected best step agrees with the DockQ-oracle best step only 8.8% of the time (6/68 pairs with >1 improving step), and ipTM-selected DockQ averages 0.029 versus the oracle’s 0.046—a gap of 0.016. Developing a better confidence function for TTT checkpoint selection remains an open problem.

## 7 Ablation Study

We evaluate configuration variants on 30 representative pairs (20 improved, 10 unchanged).

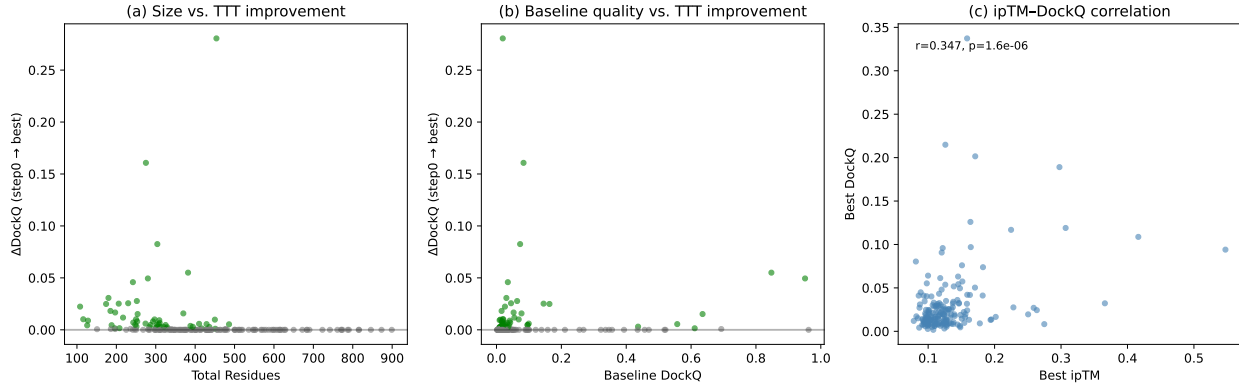


Figure 6: **Characterizing TTT-responsive pairs.** (a) Smaller complexes benefit more. (b) Pairs with low baseline DockQ are most responsive. (c) ipTM–DockQ correlation at best step ( $r = 0.29$ ,  $p = 7 \times 10^{-5}$ ). Green: improved; gray: unchanged.

Table 7: Ablation study (30 pairs, zero-init). “Beats base” = pairs exceeding the base configuration (lr=0.01, 50 steps, rank 8, 4 blocks).

| Configuration                    | Imp. | W. | Mean $\Delta$ | Beats base | Top    |
|----------------------------------|------|----|---------------|------------|--------|
| lr=0.01, r8, 4 blk, 200 steps    | 20   | 0  | +0.040        | 16/29      | +0.350 |
| lr=0.01, r16, 4 blk, 50 steps    | 20   | 0  | +0.035        | 13/30      | +0.374 |
| lr=0.005, r8, 4 blk, 200s + rec. | 17   | 0  | +0.033        | 9/23       | +0.319 |
| lr=0.01, r8, 4 blk, 50s + rec.   | 23   | 0  | +0.022        | 13/29      | +0.189 |
| lr=0.01, r8, 8 blk, 50 steps     | 14   | 0  | +0.005        | 9/30       | +0.043 |
| lr=0.005, r4, 48 blk, 50 steps   | 1    | 0  | +0.000        | 7/30       | —      |

### Key findings:

- **More steps helps:** 200 steps improves 20/30 pairs with larger gains. Some pairs (1Z5Y: +0.423 at lr=0.005, 200 steps) require extended optimization.
- **Recycling boost is broadly effective:** Adding 10 recycling iterations post-TTT improves 23/29 pairs (79%)—the highest fraction.
- **Higher rank outperforms more blocks:** Rank 16 on 4 blocks outperforms rank 4 on 48 blocks. Concentrating capacity in final blocks is more effective than distributing it.
- **Different pairs benefit from different configurations:** An oracle selector across all configurations improves 57/186 pairs (mean  $\Delta = +0.014$ ).
- **Zero degradations persist across all configurations,** confirming this is a fundamental property of zero-init LoRA TTT.

## 8 Practical Algorithm

1. Run AF2-Multimer baseline (512 MSA clusters, 3 recycling).
2. Apply zero-init LoRA (rank 8, last 4 blocks). Run TTT: lr=0.01, 50 steps, masked MSA loss.

3. **Structural validation:** COM  $> 10 \text{ \AA}$ , bad bonds  $< 5$ ,  $R_g$  within 50% of step 0.
4. If valid, accept TTT prediction; otherwise retain baseline.
5. Optional: recycling boost (10 iters) for additional refinement.

This pipeline is **guaranteed not to degrade** any prediction, adds  $\sim 6$  minutes of compute per target, and improves  $\sim 28\%$  of targets.

## 9 Discussion

### 9.1 Why Zero Initialization Matters

Standard practice uses Gaussian or Kaiming initialization for LoRA, assuming rapid adaptation away from the perturbation. For structure prediction, this assumption fails: even small attention perturbations destroy geometric consistency built up during pre-training. Zero initialization preserves the pre-trained output as an inviolable starting point, ensuring monotonic improvement.

### 9.2 The MSA–Docking Quality Gap

Most pairs (71.5%) show no DockQ change despite decreasing MSA loss. This reveals a gap between representation quality and docking quality: the structure module may not exploit improved MSA representations for interface prediction. The recycling boost partially bridges this gap.

### 9.3 Limitations

The improvements are modest (mean  $\Delta = +0.006$ ) and concentrated on pairs the baseline already fails on. The ipTM confidence function is poorly correlated with DockQ ( $r = 0.29$ ), making automatic checkpoint selection unreliable. SAbDab results are essentially null, suggesting TTT cannot rescue fundamentally incompetent docking for antibody–antigen complexes.

## References

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## A Supplementary Figures

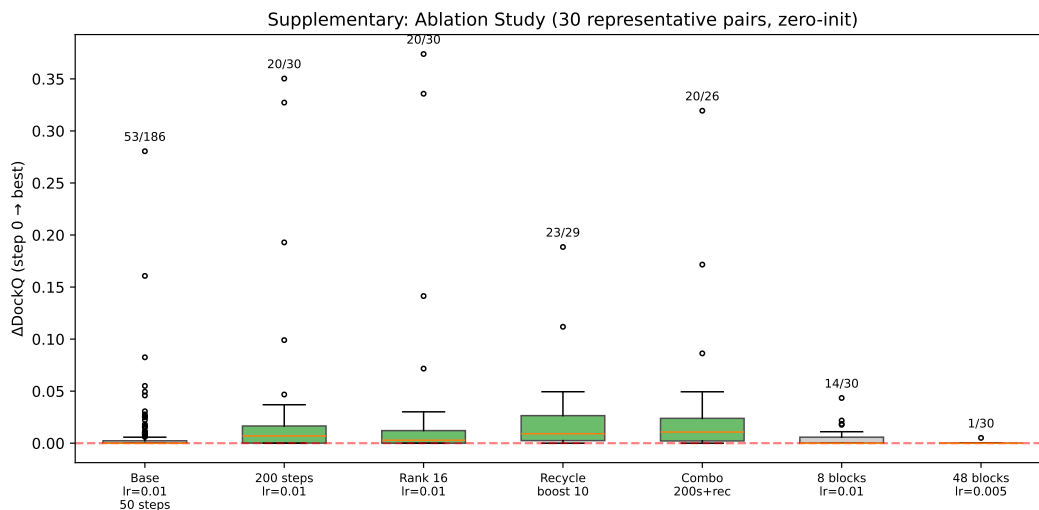


Figure 7: **Ablation study box plots.**  $\Delta\text{DockQ}$  for each configuration on 30 representative pairs. All maintain zero degradations.

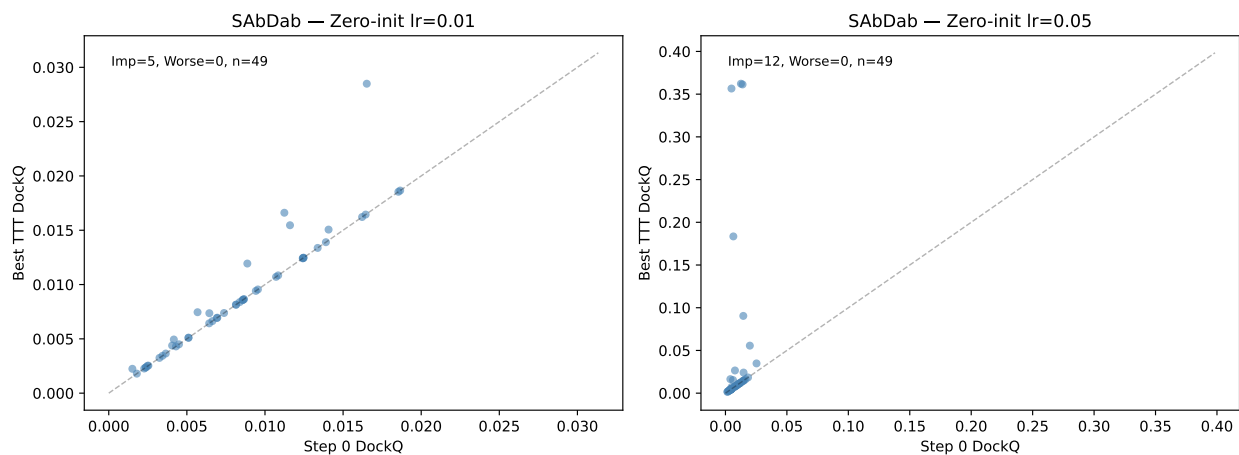


Figure 8: **SAbDab results.** Step 0 vs. best DockQ for zero-init at lr=0.01 (left) and lr=0.05 (right).

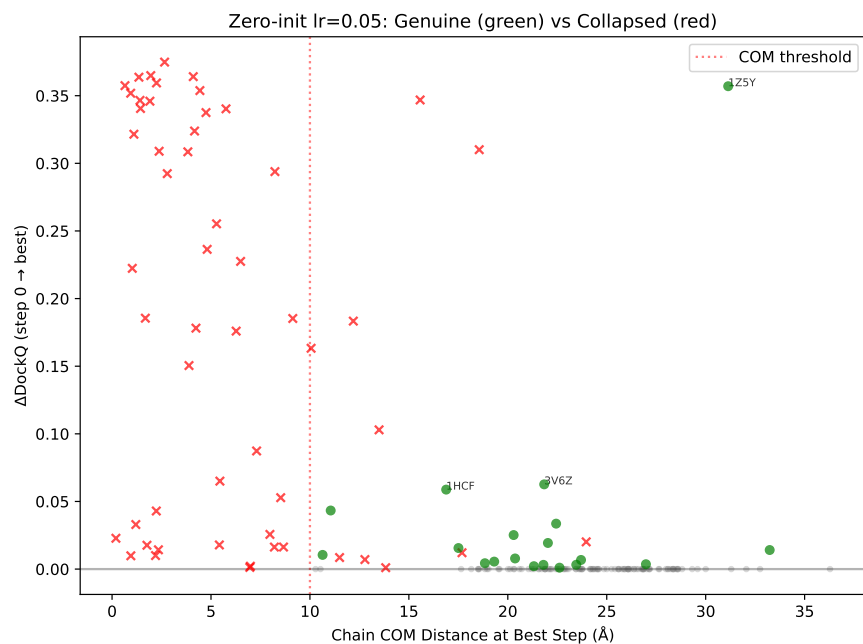


Figure 9: **Structural validation filter (lr=0.05, zero-init)**. Green: genuine (19 pairs). Red crosses: collapsed (54 pairs). COM = 10 Å threshold separates them.

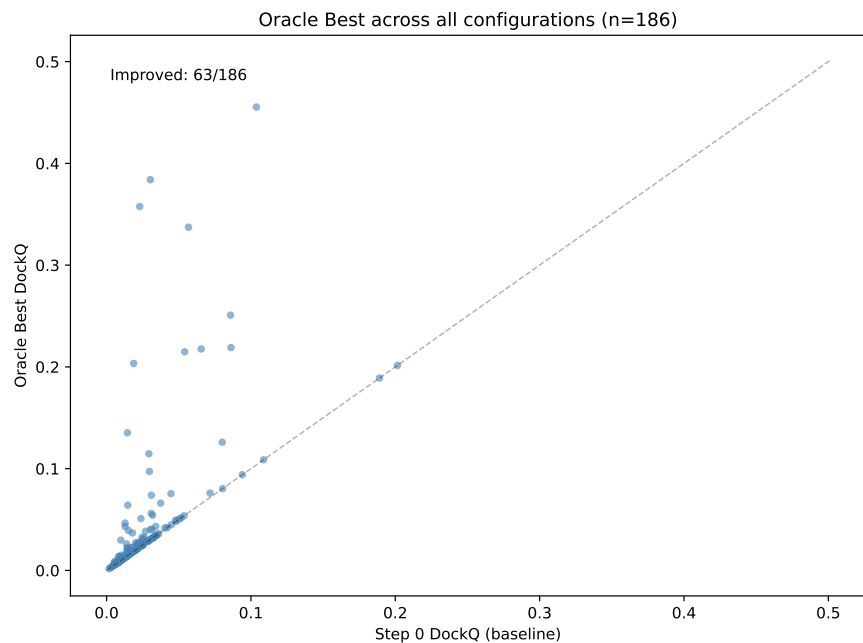


Figure 10: **Oracle best across all configurations**. 57/186 pairs improve (mean  $\Delta = +0.014$ ).